

Yixin (Stephanie) Zha

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Education

Bachelor of Applied Science in Electrical & Computer Engineering

University of Toronto

Toronto, ON

Expected May 2027

GPA: **3.94 / 4.0**

Relevant Coursework: Deep Neural Networks with PyTorch, Applied Machine Learning, Algorithms & Data Structures, Computer Organization, Digital Systems, Software Design

Skills

Languages: C++, Python, Verilog, JavaScript, Assembly, SQL, HTML/CSS

ML / Systems: PyTorch, TensorFlow Lite, MLIR, Keras, SnnTorch, NumPy, Pandas

Tools: Git, VS Code, Intel Quartus, ModelSim, SQLite, LTspice

Awards

Dean's List (2022–2025) — University of Toronto Scholar Award (2023)

ECE Outstanding Student Award (Top 2) — Intel Andy Grove Scholarship (2023)

Professional Experience

Machine Learning Compiler Engineer — Cerebras Systems

May 2025 – Aug 2026

- Developing **PyTorch frontend compiler pipelines** targeting Cerebras wafer-scale hardware using **MLIR**.
- Implementing MLIR dialects, rewrite patterns, and shape/type inference for hardware-aware lowering.
- Supporting **model bring-up and validation** for large transformer models including **GPT/LLaMA/Qwen**.
- Debugging correctness, performance, and numerical issues across PyTorch → MLIR → hardware execution.

Machine Learning Engineer Intern — Shanghai Legend Energy Ltd.

Apr 2023 – Aug 2023

- Performed time-series preprocessing and feature analysis with Pandas, improving model accuracy by **20%**.
- Evaluated and deployed 10+ ML models (LR, RF, XGBoost, VAE), improving prediction accuracy from **94% to 99.8%**.
- Built unsupervised anomaly detection models achieving **80% fault detection accuracy**.

Research Experience

Research Assistant – Edge ML for Sleep Disorder Therapy - University of Toronto

May 2025 – Aug 2026

- Designing **edge-deployed ML models** for sleep disorder treatment and closed-loop neuromodulation.
- Building and optimizing models using **TensorFlow Lite** for low-latency, low-power inference.
- Deploying inference pipelines on **Nordic nRF microcontroller boards**.

Research Assistant – Spiking Neural Networks — University of Toronto

Apr 2024 – Aug 2024

- Researched **spike-timing-dependent plasticity (STDP)** training methods for SNNs.
- Implemented SNN models using **PyTorch and SnnTorch**, achieving **99.2% accuracy** on MNIST.

Projects

University Mapping GIS Software (C++) — Designed a commuter-oriented GIS application using OpenStreetMap; led team development and Git workflow.

Real-Time Object Detection (YOLOv8) — Implemented YOLOv8 models in Python for real-time detection.

FPGA Pong Game (Verilog) — Built a two-player VGA Pong game using FSM-based control and datapath.